EagleEye: Intelligent Edge AI Surveillance

Daily Change Report and Development History

Project repository: `fyp-eagle-eye`

Report revised: 6 May 2026

Synced to latest commit `0c41c75e` on `main`

Abstract

This document records project evolution in a professional, date-ordered format. Entries follow `git log` on `main`. The **6 May 2026** section documents the most recent committed work (model-training layout and `sketchboard/` workspace). Earlier May entries summarize firmware live preview, mobile gallery tools, and related commits `7dc9cc93` and `3f6989fd`.

Contents

1 Purpose and conventions

- **Scope:** EagleEye spans ESP32-CAM firmware, a Python MQTT–cloud bridge, a React Native (Expo) mobile client, model training assets, and documentation.
- **Sources:** Commit subjects and bodies are paraphrased for clarity; hashes identify the corresponding revision where applicable.
- **Updates:** Append new dated subsections when material changes land in `main`. Recompile this `.tex` file to refresh the PDF.

2 Chronological development summary

2.1 25 October 2025

Milestone: repository bootstrap. Initial README and project upload; early ESP32-CAM bring-up on university Wi-Fi infrastructure.

2.2 21 November 2025

Milestone: on-device inference. TensorFlow Lite pipeline validated; interactive verification via DumbDisplay (tap-to-classify prototype).

2.3 26 November 2025

Maintenance. Minor commit (no extended message in history).

2.4 8 December 2025

End-to-end IoT path. Human detection with approximately 712 ms latency; intruder JPEG relayed via MQTT; Cloudinary storage with metadata URLs persisted in Firebase Realtime Database. Modular layout: camera sketch with `EagleEye_IoT.h` for connectivity.
Security hygiene. `.gitignore` extended to exclude secrets (`.env`, `secrets.h`, `serviceAccountKey.json`).

2.5 10 December 2025

Documentation. README expanded to describe system architecture and usage.

2.6 14 December 2025

Local MQTT and payload efficiency. Mosquitto-based local broker; bridge subscribed for diagnostics; ESP32 transmits raw binary JPEG (larger MQTT buffer, tuned thresholds and cooldown). Supporting assets: Mosquitto configuration, training notebooks, quality simulation utilities, dataset maintenance.

2.7 10 February 2026

Maintenance. Minor commit.

2.8 14 February 2026

Custom tiny model family. In-house training artifacts under a dedicated model directory; reference to a ∼106 ms variant and Arduino export layout.

2.9 16 February 2026

Greyscale TinyML deployment (`46847c76`). INT8 CNN at 48×48×1; RGB565 capture with software greyscale for inference; color evidence uploads unchanged; tensor arena reduced to 100 kB; simplified capture path in `EagleEye_IoT.h`.

2.10 17 February 2026

Command and control (`f803947c`). Mobile: tabbed navigation, dashboard with arm/disarm and latest alert preview, heartbeat-driven online/offline badge. Bridge: Firebase listener on `config/armed`, intrusion gating when disarmed, periodic `status/heartbeat` updates. Manual start guide updated for USB/ADB workflow.

2.11 18 February 2026

Imaging pipeline (`c79c10f7`, `719dae9b`). QVGA capture with center 240×240 crop prior to model resize—undistorted "digital zoom" for improved range; serial diagnostics updated.

2.12 6 March 2026

Academic milestone. Mid FYP presentation completed (status commit).

2.13 21 April 2026

Repository restructure (`179ef260`). Consolidated layout: `firmware/eagleeye-main`, `backend`, `mobile-app`, `model-training`, `datasets`, `docs`, `_archive`; sketch renamed to `eagleeye-main.ino`; manuals and README paths corrected.
Power-aware sensing (`d87859da`). PIR on GPIO 14, deep sleep, timed AI windows, armed-state respect, camera/Wi-Fi de-init before sleep; SD disabled to free the PIR pin.

2.14 2 May 2026 (`7dc9cc93`)

Live preview (first pass). Git message is short; the merge introduced HTTP streaming from the camera (`esp_http_server`, MJPEG at `/` on port 80), integration in `eagleeye-main.ino`, initial `LiveMonitorScreen.js` (WebView-based), dashboard navigation to Live, `pir_debug` test sketch, minor `bridge.py` edits, and test images under `backend/captures/`.

2.15 3 May 2026 (`3f6989fd`)

Live preview reliability + gallery bulk delete. Firmware: `eagleeye_grab_jpeg()` and `GET /capture`; WebSocket binary JPEG stream on port 81 (`eagleeye_ws.h`, WebSockets library); latch reset when preview ends so AI alerts work again. Mobile: WebSocket + `expo-image` live path, Android `usesCleartextTraffic`, remove unused `react-native-webview`; **Delete all** alerts with Cloudinary queue via `deletion_requests`. Documentation: this `docs/daily_changelog/` tree.

2.16 6 May 2026 (`0c41c75e`)

Model-training hygiene (`refactor(model-training)`). Experimental and superseded Python training scripts were moved into `model-training/_archive/` (e.g. COCO/Colab helpers, colour-model variants, transfer-learning and verification utilities) so the active tree is not cluttered. The single **production** training entry point for the deployed greyscale TinyML model is consolidated under `model-training/worlable_model/train_tiny_model.py` (48×48 greyscale, ESP32-oriented). A new `sketchboard/` tree provides a **scratch workspace**: copies of the working trainer, a place for firmware experiments, `notes/`, and a README describing safe day-to-day experimentation without editing production paths.

2.17 8–9 May 2026

ESP32-CAM Hard Negative Capturer. Developed a local Python Flask server (`firmware/tools/hard_neg` paired with a custom ESP32 sketch to manually capture false positives/negatives. Images are POSTed over Wi-Fi and sorted into `Humans/NonHuman` datasets. Addressed system reboots (EV-VSYNC-OVF) and Watchdog Timer timeouts during inference.

2.18 9–10 May 2026

ML Architecture Evaluation & Presentation Prep. Prepared to defend the Kaggle "Overthinkers" submission for CodeRush '26. Focused on validating the multi-model bagging/boosting pipeline, Optuna hyperparameter optimization, and adaptive blending logic.

2.19 11 May 2026

Hard Negative Retraining. Fine-tuned the custom tiny model in `sketchboard/` to integrate hard negative datasets, targeting 90%+ accuracy. Added data augmentation (flip, rotate, zoom) in `train_in_sketchboard.py`. Exported updated INT8 `.tflite` models and generated C-headers (`human_detect_model_data_augmented.h`) for firmware integration.

3 Suggested maintenance

- After each merged feature, append a dated subsection (one paragraph + bullet list if needed) and re-run `pdflatex`.
- Keep commit messages descriptive; this report is easier to maintain when subjects follow `type(scope): summary` convention.

4 Compilation

```
cd docs/daily_changelog
pdflatex EagleEye_Daily_Change_Report.tex
pdflatex EagleEye_Daily_Change_Report.tex
```